USER GUIDE

Python MongoDB Web Scraping Application

Repository URL:
https://github.com/ArbuthaDurairaj/python-mongodb-webscraping.git

1. Introduction

This project demonstrates the integration of Python web scraping and MongoDB database storage using BeautifulSoup and PyMongo.

The application collects data from public websites, processes the extracted information, and stores the results in MongoDB collections for verification and analysis.

This project was developed as part of practical database and Python web scraping exercises.

2. Project Objectives

- Perform web scraping using Python
- Extract structured data from websites
- Store scraped data in MongoDB
- Demonstrate MongoDB integration using PyMongo
- Verify inserted data using MongoDB Compass

3. Technologies Used

• Python
• MongoDB
• PyMongo
• BeautifulSoup4
• lxml
• Requests
• MongoDB Compass
• Visual Studio Code

4. Project Components

4.1 Quote Scraper
Main Files:
- index1.py
- index.py

Features:
- Multi-page scraping
- HTML parsing
- MongoDB insertion
- Automated data collection

4.2 Weather Scraper
Main File:
- weather.py

Features:
- Table extraction
- Structured data processing
- Timestamp generation
- MongoDB insertion

5. System Requirements
- Python 3.12 or later
- MongoDB Community Edition
- MongoDB Compass
- Visual Studio Code

6. Installation Instructions
Step 1 – Clone Repository
git clone https://github.com/ArbuthaDurairaj/python-mongodb-webscraping.git

Step 2 – Install Dependencies
pip install -r requirements.txt

Step 3 – Start MongoDB Service
mongodb://localhost:27017

7. Running the Application
Run Quote Scraper:
python quote_scraper/index.py

Run Weather Scraper:
python weather_scraper/weather.py

8. Database Verification

MongoDB Compass can be used to verify inserted collections and documents.

Expected collections:
- quotes
- weather

9. Project Structure

```
python-mongodb-webscraping/
├── quote_scraper/
├── weather_scraper/
├── screenshots/
├── docs/
├── requirements.txt
└── README.md
```

10. Screenshots

The repository includes screenshots demonstrating:
- Project structure
- Terminal execution
- MongoDB insert operations
- MongoDB Compass verification

11. Conclusion

This project demonstrates practical experience with Python programming, web scraping, MongoDB integration, data processing, and backend automation workflows.

Author

Arbutha Durairaj